

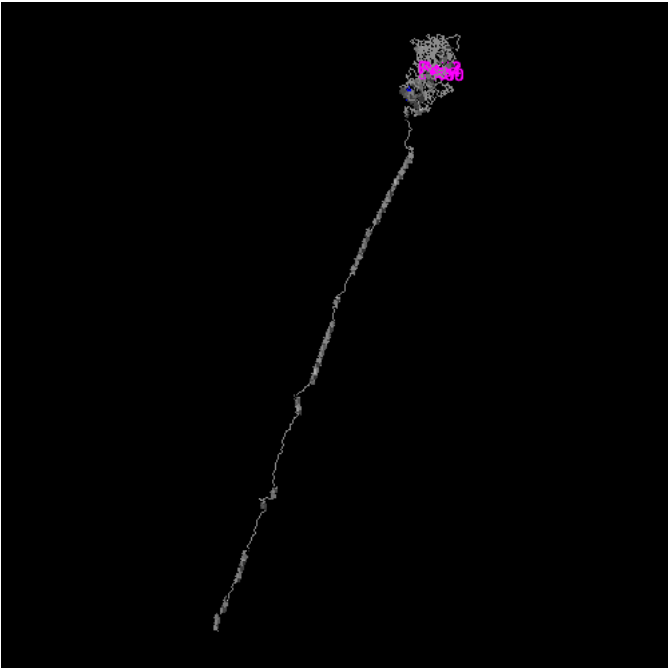
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COACH results for job CH2099421454

[Download the predicted 3D model](#) (This is a hard target for structure modeling. The structure was predicted by SPARKS-X (SPX) with template [7wqxΔ](#). z-score=11.120 (cutoff=6.9).)

COACH Results

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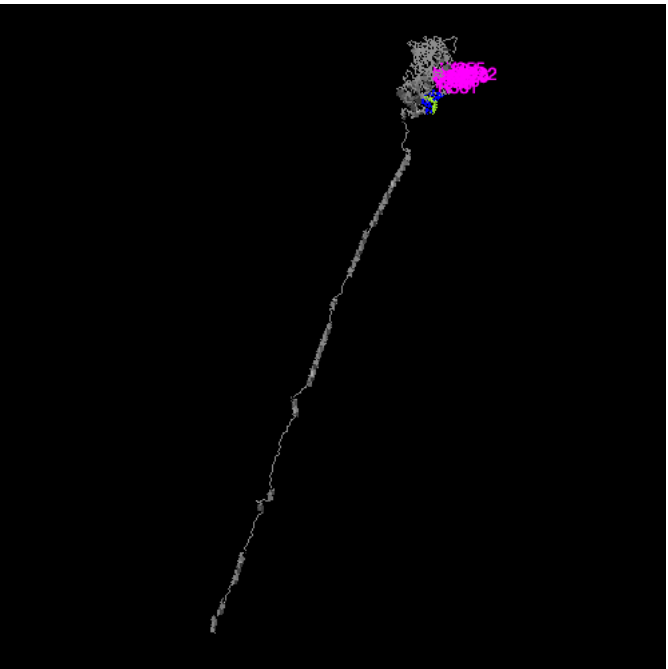
☐ Spin On/Off

Click to view	Rank	C-score	Cluster size	PDB Hit	Lig Name	Download Complex	Consensus Binding Residues
☒	1	0.27	15	1w52X	CA	Rep. Mult	457,460,461,462,465
☐	2	0.09	4	1lpbB	MUP	Rep. Mult	346,347,421,422,423,448,450,485,533
☐	3	0.08	7	1ethA	C8E	Rep. Mult	347,348,349,422,485,522,526,530,533
☐	4	0.06	3	1tg7A	MAN	Rep. Mult	396,436
☐	5	0.06	5	1ethC	C8E	Rep. Mult	345,346,347,348,349,355,381,421,422,522,533
☐	6	0.02	2	1w52X	DDQ	Rep. Mult	348,349,384,485,533
☐	7	0.02	2	2oxeB	MAN	Rep. Mult	633,635
☐	8	0.01	1	N/A	N/A	N/A	358,361
☐	9	0.01	1	N/A	N/A	N/A	366,441,547,580,747,854,901,902,905,927
☐	10	0.01	1	1bruP	1NB	N/A	763,900,901,902,903,904,905,924,925

- [Download](#) the residue-specific binding probability, which is estimated by SVM.
[Download](#) the predicted bound ligands and detailed prediction summary.
[Download](#) the templates clustering results.
- (a) **C-score** is the confidence score of the prediction. C-score ranges [0-1], where a higher score indicates a more reliable prediction.
(b) **Cluster size** is the total number of templates in a cluster.
(c) **Lig Name** is the name of possible binding ligand. Click the ligand name to view its information in the [database](#).
(d) **Rep** is a single complex structure with the most representative ligand in the cluster, i.e., the one listed in the **Name** column.
Mult is the complex structures with all potential binding ligands in the cluster.

TM-SITE Results

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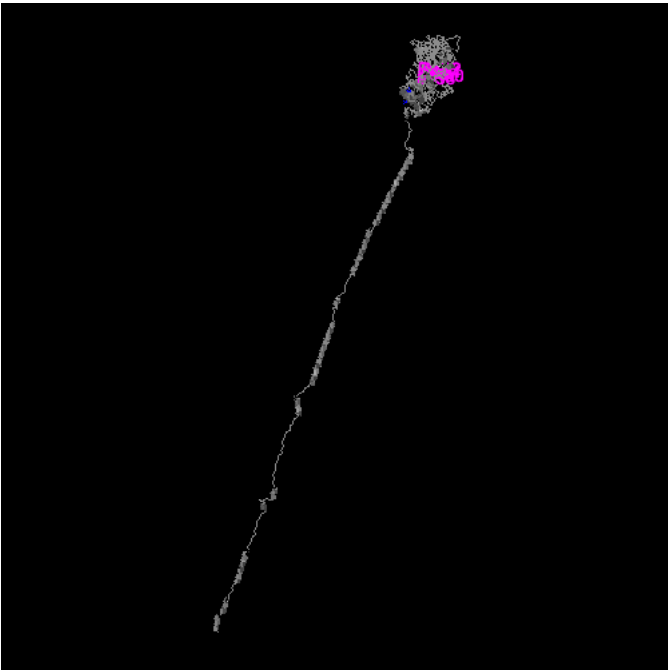
☐ Spin On/Off

Click to view	Rank	C-score ^a	Cluster size ^b	Rep Templ ^c	Mult Templ ^d	Ligands ^e	Predicted binding site
☐	1	0.34	4	1ethC_BS05	C8E	Download C8E (4)	345,346,347,348,349,355,381,421,422,423,448,450,485,533
☐	2	0.16	1	1lpbB_BS02	MUP	Download MUP (1)	346,347,421,422,423,448,450,485,533

- (a) **C-score** is the confidence score of predicted binding site. C-score ranges [0-1], where a higher score indicates a more reliable prediction.
(b) **Cluster size** is the number of templates in a cluster.
(c) **Rep Templ** presents a representative ligand-protein template from a cluster. The template names are #_(Ligand ID). Click the corresponding template to download the structure.
(d) **Mult Templ** provides all ligand-protein complex structures in a cluster, where each ligand is separated by a semicolon.
(e) **Ligands** lists all ligands in a cluster. The numbers in the parentheses are the appearing times of the corresponding ligand name to visualize its detailed information in BioLiP.

S-SITE Results

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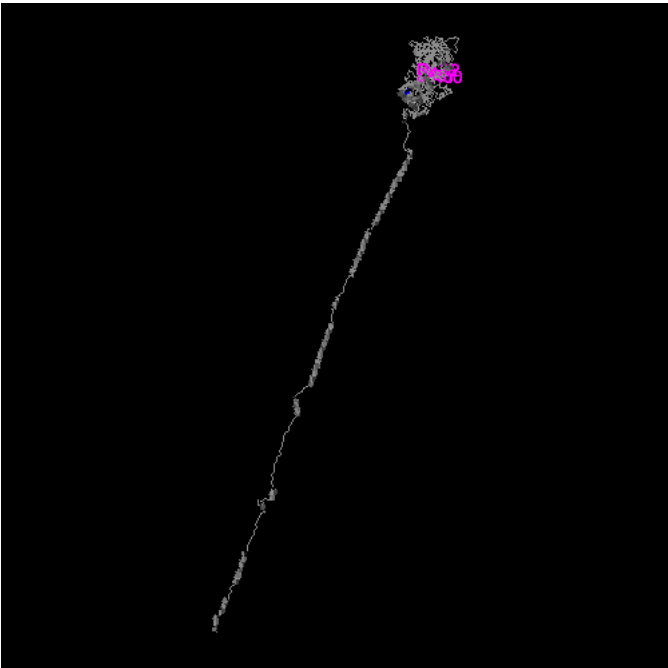
☐ Spin On/Off

Click to view	Rank	C-score ^a	Cluster size ^b	Templates ^c	Ligands ^d	Predicted binding site
☐	1	0.20	5	List	CA (5)	457,458,460,461,462,465
☐	2	0.12	2	List	PLC (1), MUP (1)	346,347,348,384,421,422,423,448,450,4
☐	3	0.11	2	List	IOD (1), 1NB (1)	746,747,763,900,901,902,903,904,905,9
☐	4	0.08	1	List	DMF (1)	730,845,864

- (a) **C-score** is the confidence score of predicted binding site. C-score ranges [0-1], where a higher score indicates a more reliable prediction.
- (b) **Cluster size** is the number of templates in a cluster.
- (c) **Templates** presents the list of templates in a cluster. The template names are: (PDBIDchain)_(BioLiP corresponding template) to search the BioLiP database. When the number of templates is >5, click "show all templates in the cluster".
- (d) **Ligands** lists ligands in a cluster. The numbers in the parentheses are the appearing times for the corresponding ligand name to visualize its detailed information in BioLiP.

COFACTOR Results

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☐ Spin On/Off

Click to view	Rank	C-score ^a	PDB Hit	TM-score	RMSD ^b	IDEN ^c	Cov ^d	BS-score ^e	Lig. Name	Download Complex	Predicted binding site
☐	1	0.47	1hplA	0.418	2.93	0.810	0.443	1.86	CA	Download	457,460,462,465
☐	2	0.34	1ethA	0.440	2.54	0.736	0.461	1.94	BMA	Download	433,434
☐	3	0.33	1lpbB	0.438	2.46	0.697	0.459	1.99	MUP	Download	346,347,421,422
☐	4	0.15	1ethC	0.440	2.54	0.736	0.461	1.23	C8E	Download	345,346,347,348
☐	5	0.14	1w52X	0.418	3.05	0.527	0.448	1.12	DDQ	Download	348,349,384,485

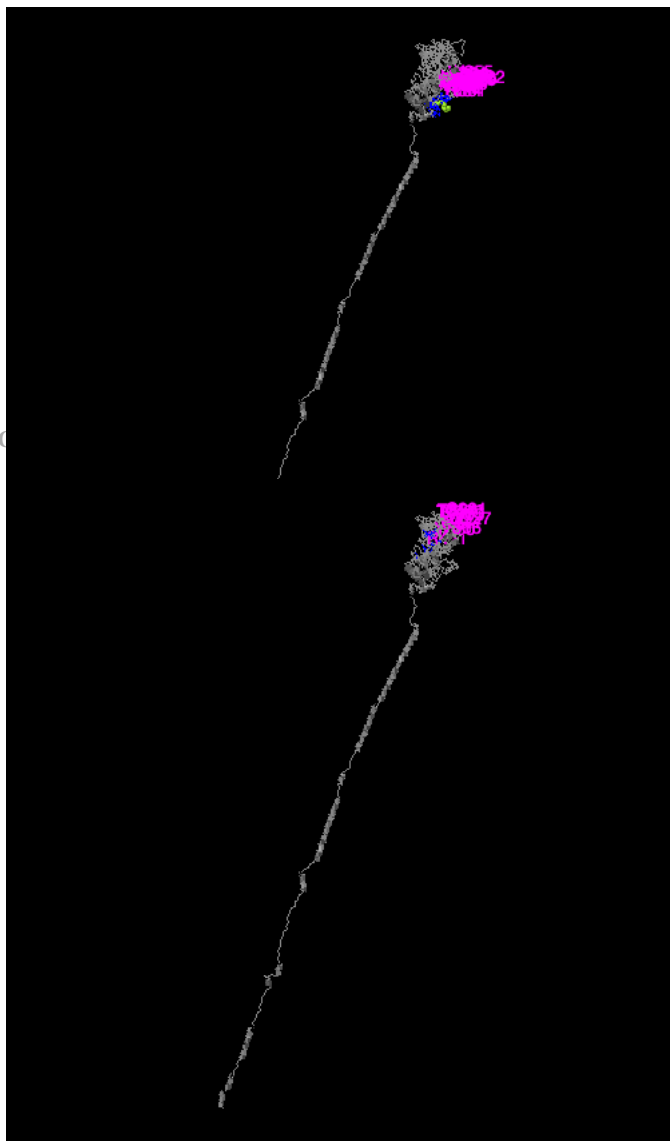
- (a) **C-score** is the confidence score of predicted binding site. C-score values range in between [0-1]; where a higher score indicates a more reliable prediction.
- (b) **RMSD** the RMSD between residues that are structurally aligned by TM-align.
- (c) **IDEN** is the percentage sequence identity in the structurally aligned region.
- (d) **Cov** represents the coverage of global structural alignment and is equal to the number of structurally aligned residues of the query protein.
- (e) **BS-score** is a measure of local similarity (sequence & structure) between template binding site and query structure. Based on large scale benchmarking analysis, we have observed that a BS-score >1 reflects a high similarity between the predicted and template binding site.

FINDSITE Results

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Click to view	Rank	C-score ^a	Cluster size ^b	Rep Templ ^c	Multi Templ ^d	Ligands ^e	Predicted binding site
☐	1	0.38	6	1ETHA02	List	download	345,346,347,348,349,355,381,384,421,4
☐	2	0.12	2	1LPAB00	List	download	350,353,354,381,522
☐	3	0.06	1	2OXEA00	List	download	546,604,606,633,634,636,660,661
☐	4	0.06	1	1W52A00	List	download	508,524,525,528,529
☐	5	0.06	1	1LPBB03	List	download	518,519,523,527,535,538,602,603,604,6

- (a) **C-score** is the confidence score of predicted binding site. C-score values range in between [0-1]; where a higher score indicates a more reliable prediction.
- (b) **Cluster size** is the number of templates in a cluster.
- (c) **Rep Templ** presents a representative ligand-protein template from a cluster. The template names are: (PDBIDchain)_(BioLiP corresponding template) by FINDSITE as 2-digit numbers. Click the corresponding template to download the structure.
- (d) **Multi Templ** provides the list of all templates in a cluster.

(e) **Ligands** presents the structures of all ligands in a cluster.
☐ Spin On/Off

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Click to view Rank C-score			Predicted binding residues
☐	1	0.52	366,441,547,580,747,854,901,902,905,927
☐	2	0.12	358,361

Please cite the following articles when you use the COACH server:

1. Jianyi Yang, Ambrish Roy and Yang Zhang, *Protein-ligand binding site recognition using complementary binding-specific substructure comparison and sequence profile alignment*, Bioinformatics, 29: 2588-2595 (2013) ([download the PDF file](#))
2. Jianyi Yang, Ambrish Roy and Yang Zhang, *BioLiP: a semi-manually curated database for biologically relevant ligand-protein interactions.*, Nucleic Acids Research, 41: D1096-D1103 (2013) ([download the PDF file](#)).